温州肯恩大学
WENZHOU-KEAN UNIVERSITY

GMBA 5885 W03
Strategic Mgt Global Info Sys

Managing Global Information Systems

Dawei Chen

22 Nov 2025

Kick-Off Discussion

- Warm-Up Questions
  - What challenges do companies face when operating in many countries?
  - How do cultural, legal, and technological differences shape global IT needs?
  - Should global firms standardize systems worldwide, or adapt to each local market?
  - What examples (e.g., Apple, McDonald's, TikTok, Airbnb) illustrate global IS complexity?
- Think About
  - How do companies balance global efficiency with local responsiveness?
  - What happens when different regions use incompatible systems?
  - Why is managing data, security, and connectivity harder across borders?
- Goal for Today
  - Understand how global strategy, organizational structure, and technology decisions come together to build effective global information systems.

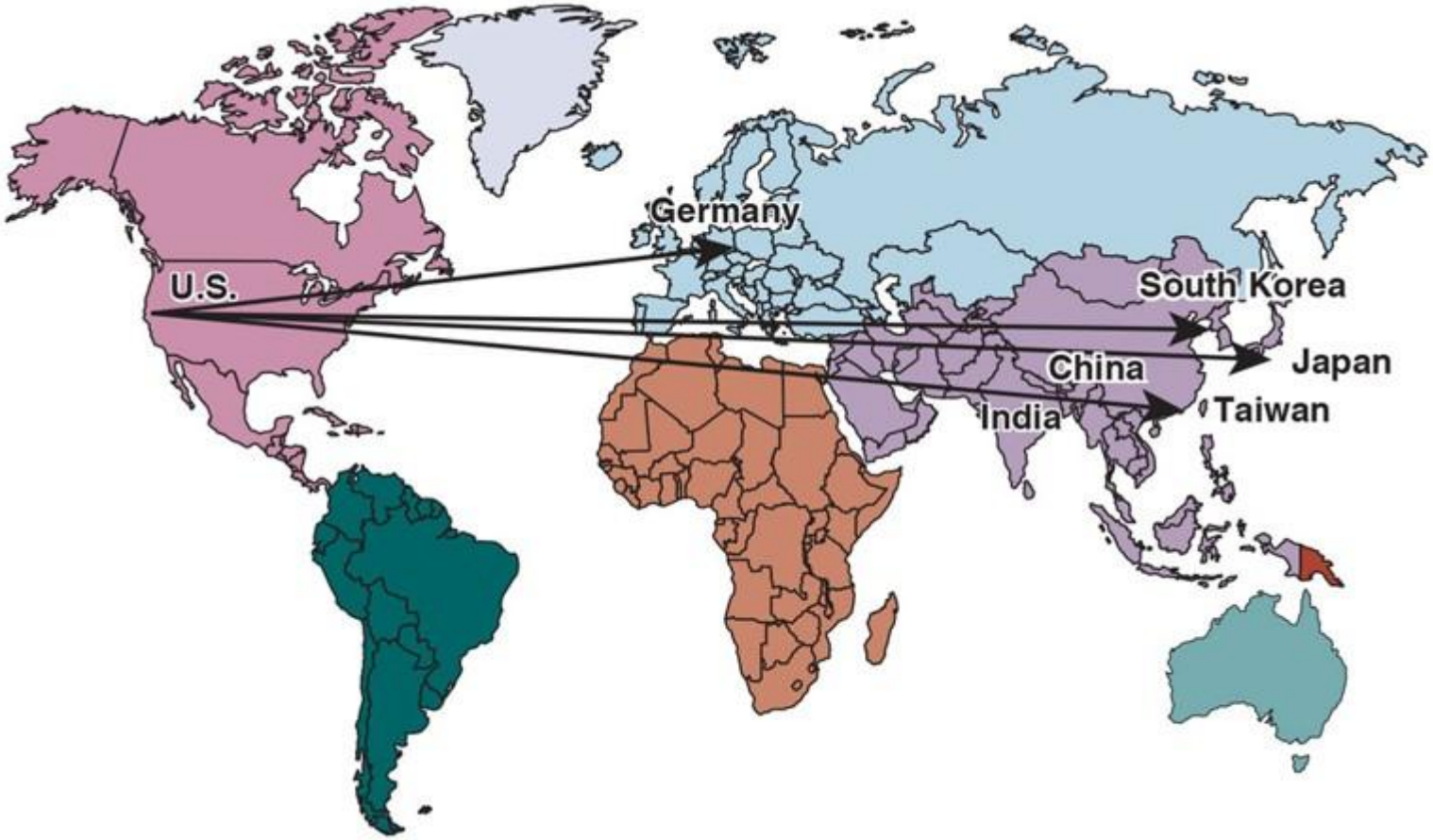
Learning Objective

1) Identify factors affecting global information systems
2) Outline effective global IS architecture
3) Discuss strategies and organizational structures for global business and IS
4) Describe management challenges in developing global IS
5) Discuss solutions to address these challenges
6) Discuss technology issues and alternatives for global IS development

Factors Impacting Development of Global IS

- Global economic system and global world order driven by advanced networks and information systems
- The growth of international trade has radically altered domestic economies around the globe
- For example, production of many high-end electronic products is divided out among many different countries
  - For example: Apple iPhone’s global supply chain

Apple iPhone’s Global Supply Chain



Global Business Drivers

General Cultural Factors	Specific Business Factors
Global information, communication, and transportation technologies	Global markets
Development of global culture	Global production and operations
Global knowledge base	Global coordination
	Global workforce
	Global economies of scale

Global Business Challenge

Global Factors	Specific Factors
Cultural particularism: nationalistic, religious, language differences	Standards: different EDI, email, and telecommunications standards
Social expectations: brand-name expectations, work hours	Reliability: telecommunications networks not uniformly reliable
Political and legal: transborder data/privacy laws, commercial regulations	Speed: varying data transfer speeds, often slower than the U.S.
Deglobalization and regionalization trends driven by shocks (e.g., Covid-19)	Personnel: shortages of skilled consultants

Develop Global IS Architecture

Global Strategies

Four Global Business Strategies

- Domestic Exporter
  - Highly centralized; home country controls production, finance, marketing
  - Example: Leatherman (U.S.-designed and manufactured)
- Multinational
  - Centralized financial control; decentralized production/marketing– Local units adapt products to local markets
  - Example: Hormel Foods

Global Strategies

Four Global Business Strategies

- Franchiser
  - Product designed/controlled at home; production/HR/marketing local
  - Example: McDonald’s, Starbucks
- Transnational
  - No single national HQ; regional/global hubs
  - Integrated global operations, optimized worldwide
  - Example: Unilever, Nestlé
- Key Insight: Firms must balance global efficiency (standardization, scale) with local responsiveness (adaptation, regulations). Information systems enable both.

Global Strategy and Systems Configurations

SYSTEMS CONFIGURATION	Strategy			
	Domestic Exporter	Multinational	Franchiser	Transnational
Centralized	X			
Duplicated			X	
Decentralized	x	X	x	
Networked		x		X

Development and Operations

Reorganizing the Business for Global Operations

Key Principles for Global Structure

- Align activities by comparative advantage
  - Place marketing, production, finance, HR, and IS where they perform best in cost, impact, and reliability
  - Build supply chain resiliency into global production decisions
- Develop systems units at all levels
  - Local (host country): support local operations and requirements
  - Regional: manage telecom and cross-border systems within major regions
  - Global (transnational): link regions and coordinate global systems development
- Create a global CIO role
  - A single authority to lead global IS development and integration
- Success Factor
  - A management team capable of balancing global opportunities with risks and guiding the development of effective international systems.

Management Challenges in Developing Global IS

- Agreeing on common user requirements
- Introducing changes in business processes
- Coordinating applications development
- Coordinating software releases
- Encouraging local users to support global information systems

A Typical Scenario: Disorganization on a Global Scale

Current Situation

- Based in U.S., operating in Europe, expanding into Asia
- Regional units built **independent systems** for production, marketing, HR, and sales
- Only **financial reporting** is centralized
- Result: fragmented ERP/MRP systems, inconsistent hardware/software, unreliable cross-country communication

A Typical Scenario: Disorganization on a Global Scale

Key Problems

- No common user requirements
- Local divisions resist standardization
- U.S. systems group focused only on domestic needs
- Local managers rewarded for local—not global—performance
- Difficult to coordinate global development and gain user buy-in

A Typical Scenario: Disorganization on a Global Scale

Recommendation Focus

- Move toward **transnational strategy** with unified global IS architecture
- Standardize core platforms; coordinate global and regional systems development
- Build incentives, governance, and communication to support global integration

Reorganizing the Business

How should a firm organize itself for doing business on a global scale?

- To develop a global company and information systems support structure, a firm needs to follow these principles
  1. Organize value-adding activities along lines of comparative advantage
  2. Develop and operate systems units at each level of corporate activity: regional, national, and international
  3. Establish a single office responsible for the development of global information systems

Management Solutions for Global IS Development

Step 1: Focus on Core Systems

- Not all systems must be global
- Identify core business processes (e.g., order fulfillment, supply chain, HR)
- Centralize only the systems that support these core processes

Step 2: Analyze and Prioritize Processes

- Map end-to-end processes (orders, suppliers, shipping, etc.)
- Identify centers of excellence across regions
- Decide which processes should be global, regional, or local

Examples of Core, Regional, and Local Business Processes

Type of Business Process	Examples
Core business processes	Managing global supply chains Maintaining a global brand image Establishing global quality standards
Regional business processes	Operating regional distribution centers Running a regional marketing campaign Evaluating regional sales district performance Providing region-specific customer service
Local business processes	Taking customer orders Customizing food product labels for different languages and local tastes Training local employees Hiring local office staff

Management Solutions for Global IS Development

Step 3: Keep the Global System List Small

- Choose only a few critical transnational systems
- Minimize resistance by allowing local/peripheral systems to vary

Step 4: Use an Evolutionary Approach

- Avoid piecemeal or “grand design” strategies
- Build global capabilities incrementally (“salami strategy”) with a 5-year vision

Step 5: Make Benefits Clear

- Better global coordination & crisis response
- Improved production, operations, and supply chain resiliency
- Global customer base → economies of scale
- Optimized use of corporate funds across regions

Implementing Management Solutions

Management challenges in developing global information challenges

- Agreeing on common user requirements
- Introducing changes in business processes
- Coordinating applications development
- Coordinating software releases
- Encouraging local users to support global information systems
  - Overall tactic for dealing with resistant local units in a transnational company is co-optation
  - Bringing the opposition into the process of designing and implementing the solution without giving up control

Technology Challenges in Global IS

Key Issues

- Hard to standardize hardware, software, and networks across countries
- Different platforms (Windows, Linux, Unix, local systems)
- Incompatible collaboration tools across regions
- Variable Internet quality despite global adoption

Need

- Global systems integration + consistent connectivity.

Computing Platforms & System Integration

Challenges

- Integrating many legacy systems across countries
- Ensuring shared data standards (fiscal year, accounting terms, interfaces)
- ERP/SCM systems may conflict with local languages, culture, regulations
- Some countries lack the technical skills to manage complex enterprise apps

Key Tools

- ERP, SCM, EDI, collaboration platforms
- Global standards for data, interfaces, and architectures

Connectivity & Global Networks

- Connectivity Requirements
  - Unified global network supporting voice, data, video
  - Public Internet ≠ guaranteed reliability or security
  - Firms rely on private networks or VPNs for sensitive data
- Global Constraints
  - Poor Internet infrastructure in parts of Africa, Middle East, Asia-Pacific
  - Limited bandwidth, unstable power grids
  - High cost of Internet access in developing regions
- Options
  - Global intranets, extranets
  - VPNs (lower cost, but variable performance)

Software Localization

Key Challenges

- Integrating new global systems with diverse legacy systems
- Building interfaces for old-to-new system connections (costly, complex)
- Creating software usable across countries with different processes and data definitions

Human Interface Issues

- Interfaces must be simple, intuitive, and quick to learn
- Cannot assume a single language—must support **multilingual UI** and local conventions
- Localization required as systems move beyond knowledge workers to broader staff

Role of AI

- AI tools increasingly automate translation and localization, reducing time and cost